WEEK-5
POWER BI INTERNSHIP

TASK 1 – POWER BI INTRODUCTION & DATA IMPORT

Aim

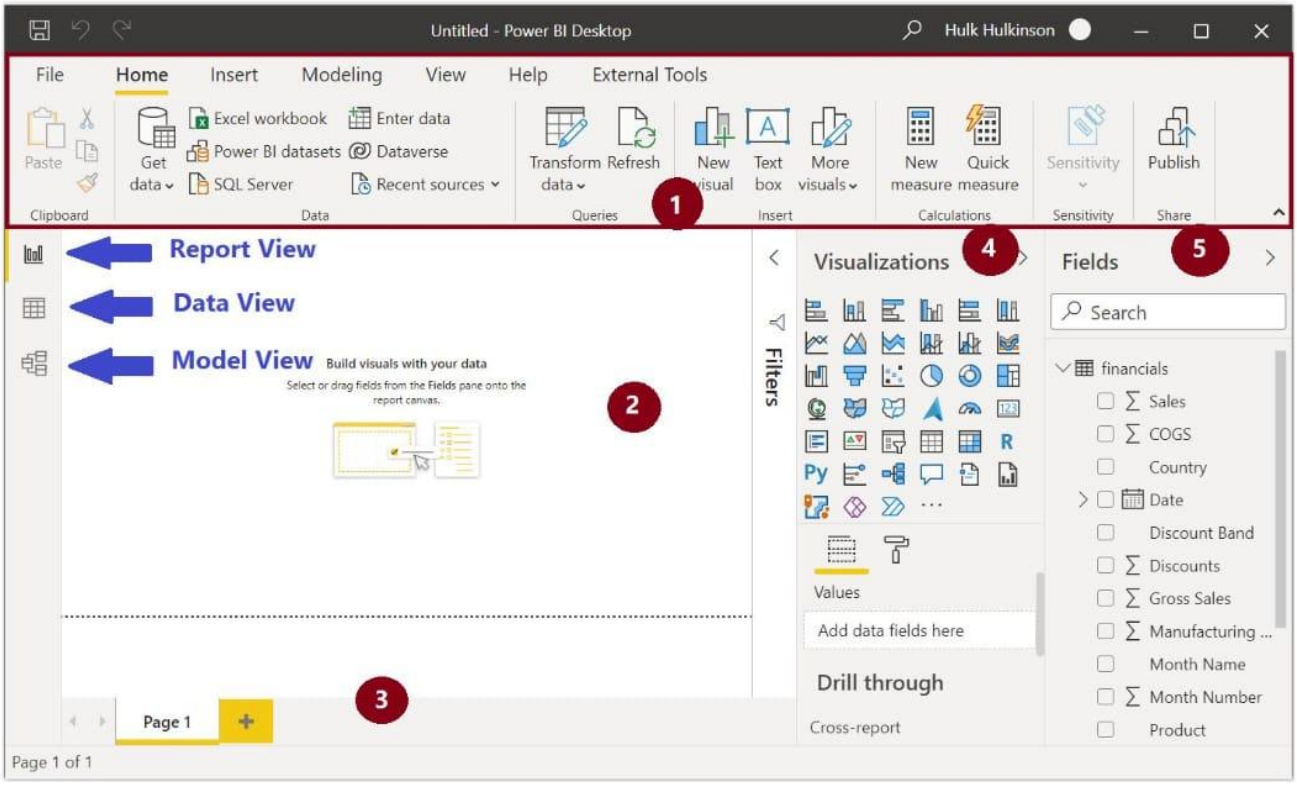
To understand Power BI and import the Excel dataset.

Procedure

1. Open Power BI Desktop.
2. Select Home → Get Data → Excel Workbook.
3. Browse and select the internship workbook.
4. In Navigator, select Sales, Products and Customers.
5. Select Load.
6. Open Data view and verify the tables and column data types.

Result

The Sales, Products and Customers tables are imported and ready for preparation. Successfully explored Power BI Desktop and imported the Excel dataset containing three tables: Sales, Products and Customers.



TASK 2 – DATA CLEANING & PREPARATION

Aim

To remove duplicate records, handle missing values and assign correct data types.

Power Query Procedure

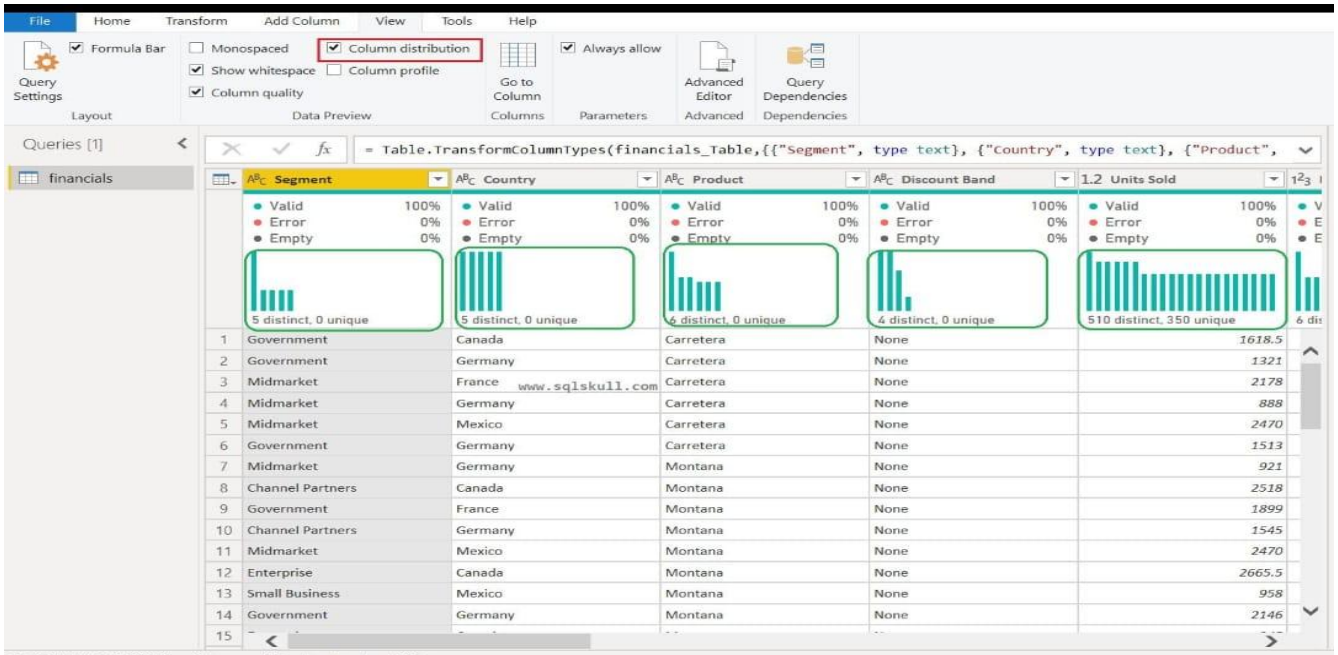
1. Select Home → Transform Data.

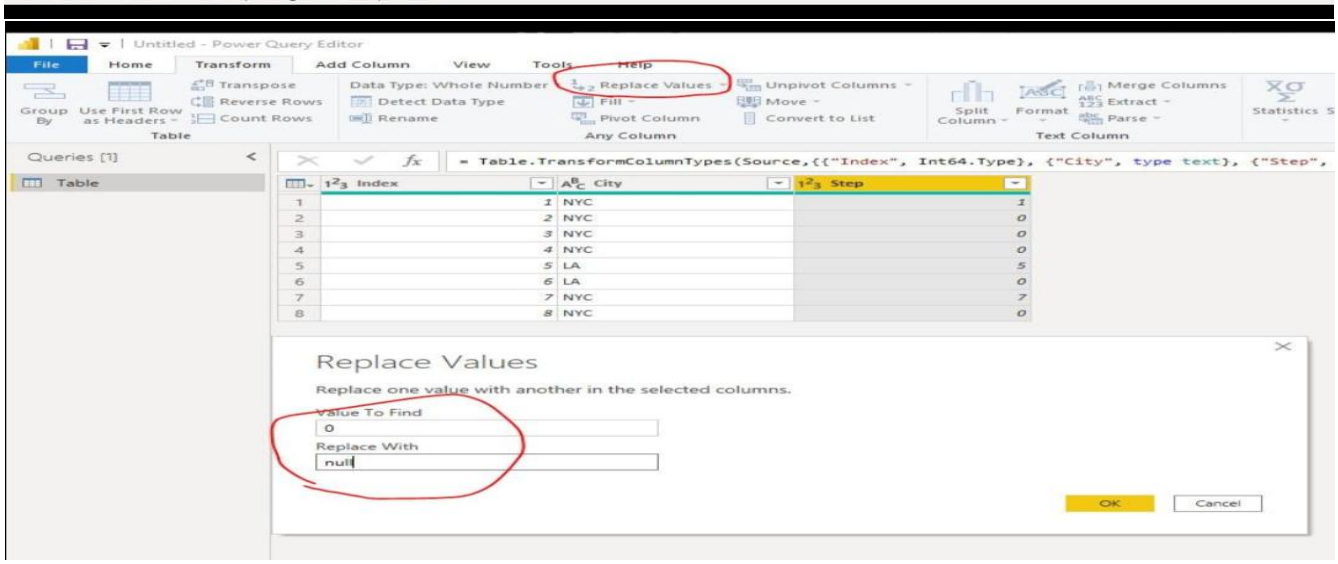
2. Select the Sales query.
3. Select OrderID and choose Remove Rows → Remove Duplicates.
4. Replace the blank/null Region value with “Unknown”.
5. Set OrderDate to Date, Quantity to Whole Number, and UnitPrice/SalesAmount to Decimal or Currency.
6. Check Column Quality for errors and empty values.
7. Select Close & Apply.

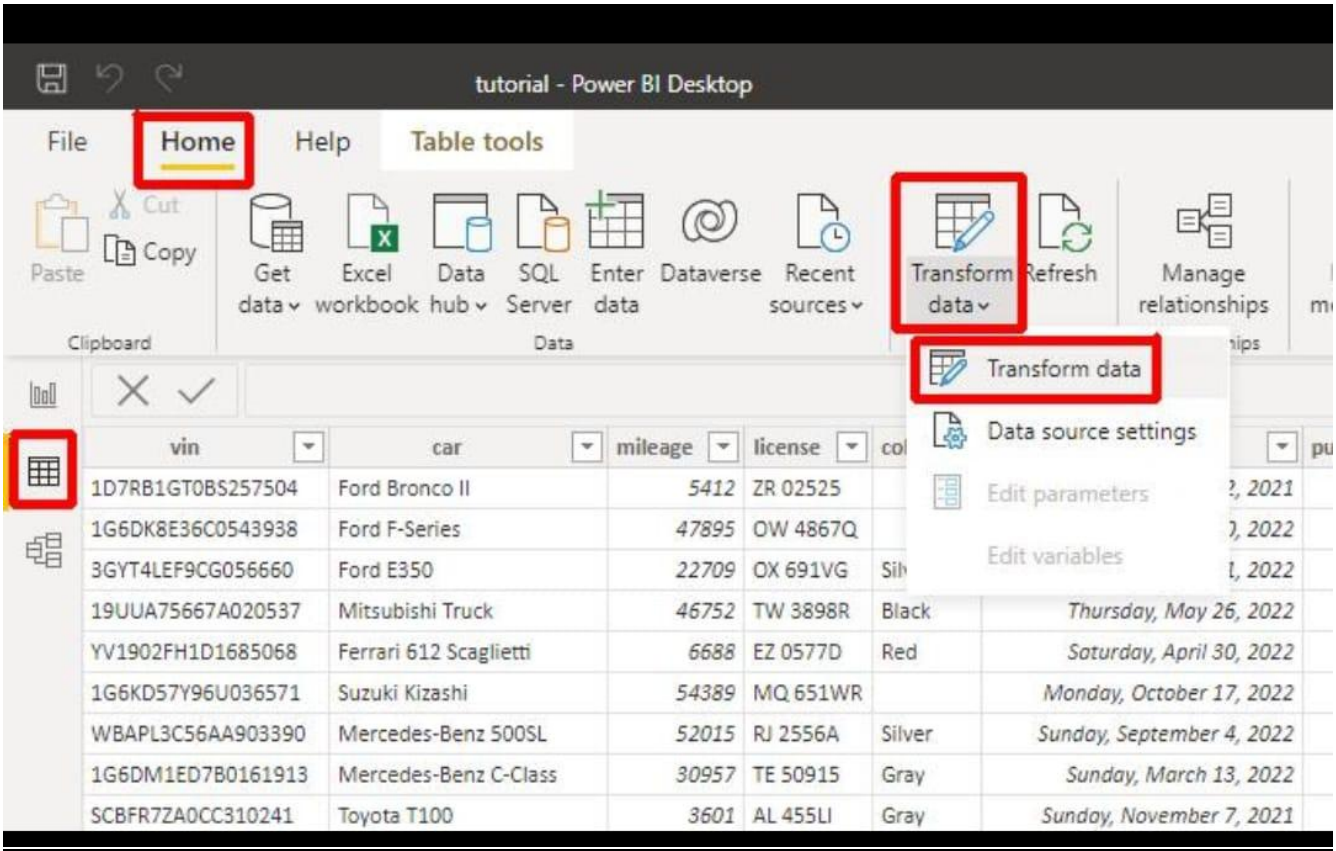
Check	Before	After
Sales rows	22	21
Duplicate transaction	Present	Removed
Blank Region	1 blank	Unknown

Result

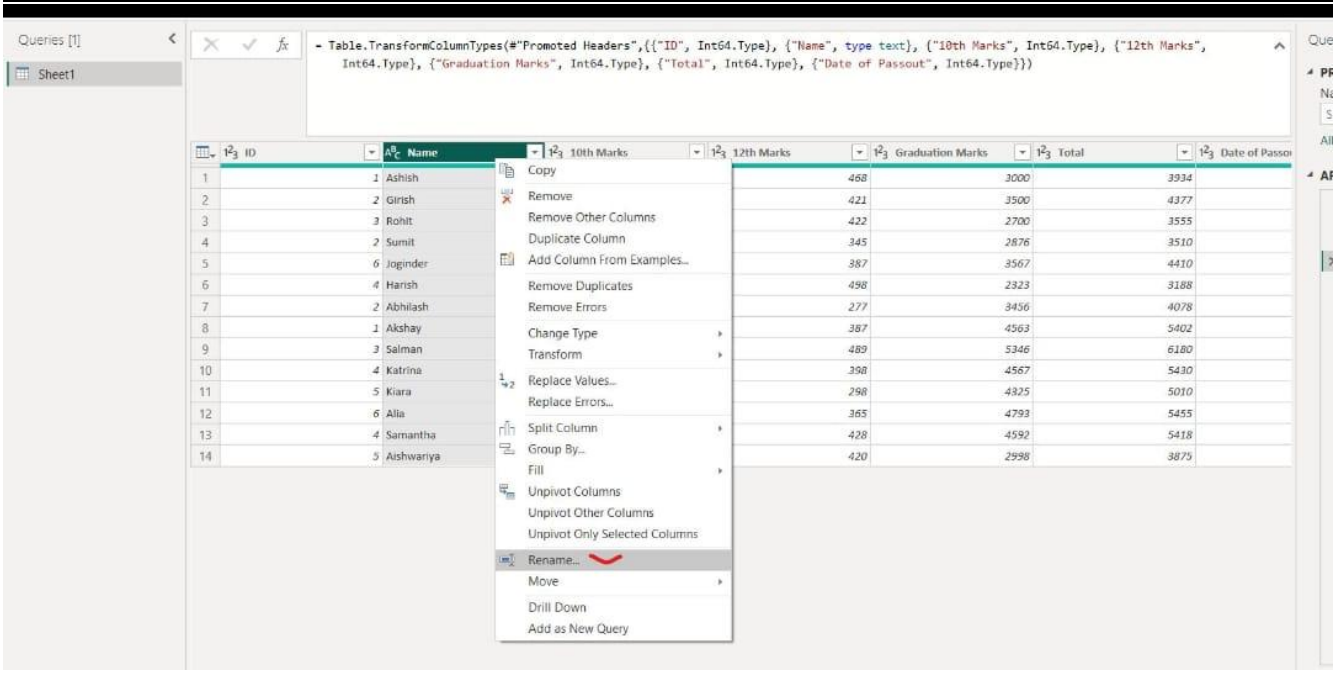
The cleaned Sales table contains 21 records and has no duplicate transaction. The missing Region value is represented as Unknown.Successfully cleaned and prepared the dataset using Power Query Editor. Duplicate records were removed, missing values were handled, column names were improved and correct data types were applied. The final Sales table contains 21 records and is ready for further analysis.







vin	car	mileage	license	col	
1D7RB1GT0BS257504	Ford Bronco II	5412	ZR 02525		, 2021
1G6DK8E36C0543938	Ford F-Series	47895	OW 4867Q		, 2022
3GYT4LEF9CG056660	Ford E350	22709	OX 691VG	Sil	, 2022
19UUA75667A020537	Mitsubishi Truck	46752	TW 3898R	Black	Thursday, May 26, 2022
YV1902FH1D1685068	Ferrari 612 Scaglietti	6688	EZ 0577D	Red	Saturday, April 30, 2022
1G6KD57Y96U036571	Suzuki Kizashi	54389	MQ 651WR		Monday, October 17, 2022
WBAPL3C56AA903390	Mercedes-Benz 500SL	52015	RJ 2556A	Silver	Sunday, September 4, 2022
1G6DM1ED7B0161913	Mercedes-Benz C-Class	30957	TE 50915	Gray	Sunday, March 13, 2022
SCBFR7ZA0CC310241	Toyota T100	3601	AL 455LI	Gray	Sunday, November 7, 2021



TASK 3 – DATA MODELING & RELATIONSHIPS

Aim

To connect the Sales table to Products and Customers using matching ID columns.

Procedure

1. Open Model view.
2. Connect Products[ProductID] to Sales[ProductID].
3. Set cardinality to One-to-many (1:*).
4. Connect Customers[CustomerID] to Sales[CustomerID].

5. Set cardinality to One-to-many (1:*).
6. Keep relationships active with single-direction filtering from lookup tables to Sales.

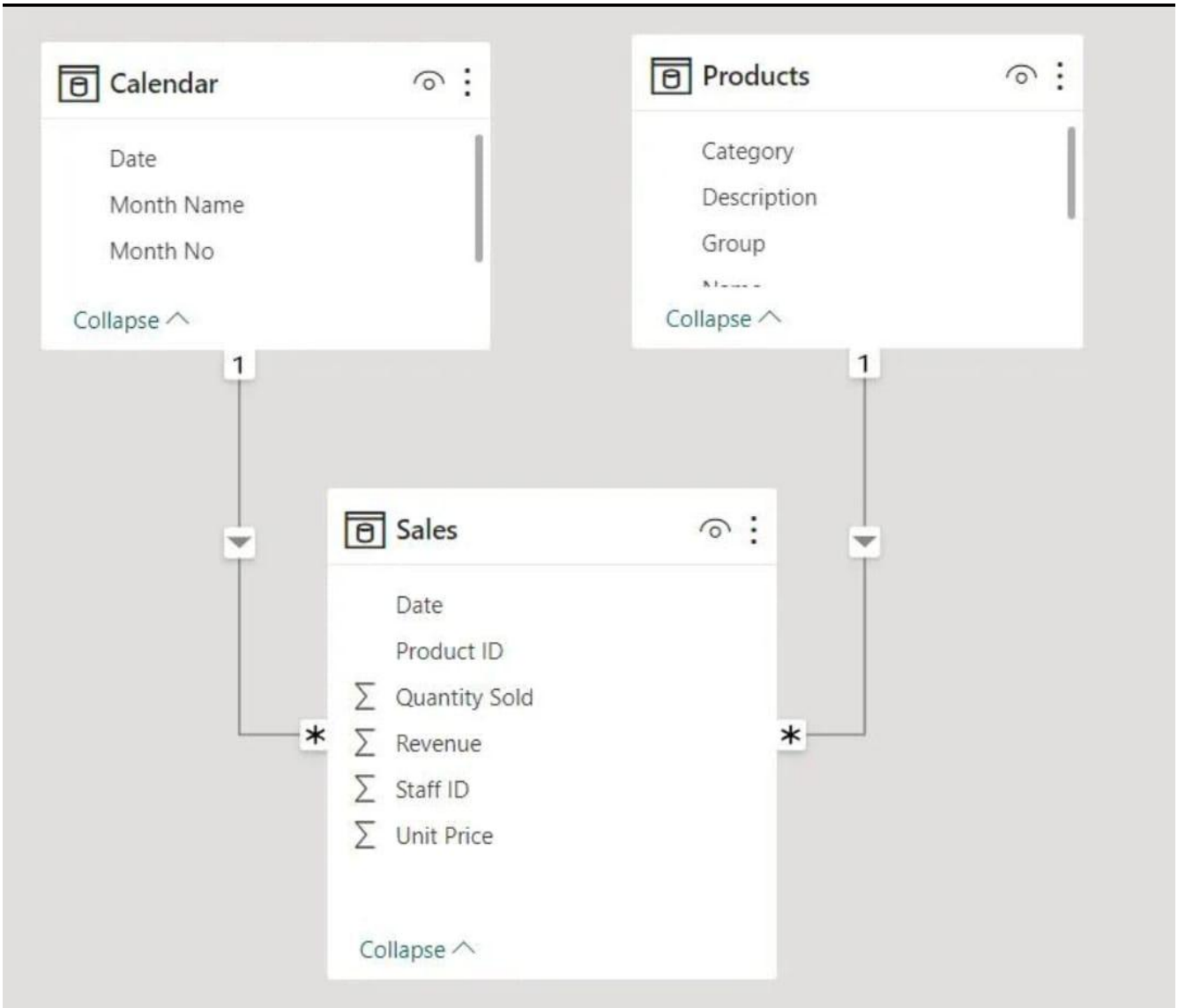
One-side table	Many-side table	Cardinality
Products[ProductID]	Sales[ProductID]	1:*
Customers[CustomerID]	Sales[CustomerID]	1:*

Illustrative model: PRODUCTS (1) ■■■■■ (*) SALES (*) ■■■■■ (1) CUSTOMERS

Screenshot note: This is a text illustration, not an actual Power BI screenshot. Insert your own Model view screenshot if your supervisor requires application screenshots.

Result

The Sales table is connected to Products and Customers through their matching ID columns. Successfully created a basic data model in Power BI using three tables. Established two active Many-to-One relationships and prepared the model for DAX calculations and visualizations.



TASK 4 – DAX CALCULATED COLUMNS & MEASURES

Aim

To create calculations that summarize sales performance.

Procedure

1. Select the Sales table.
2. Choose Modeling → New measure for measures.
3. Enter each DAX formula in the formula bar and press Enter.
4. Choose Modeling → New column for Calculated Amount.
5. Format currency measures as Currency.

DAX Formulas

```
Total Sales = SUM(Sales[SalesAmount])


Average Sales = AVERAGE(Sales[SalesAmount])


Total Records = COUNTROWS(Sales)

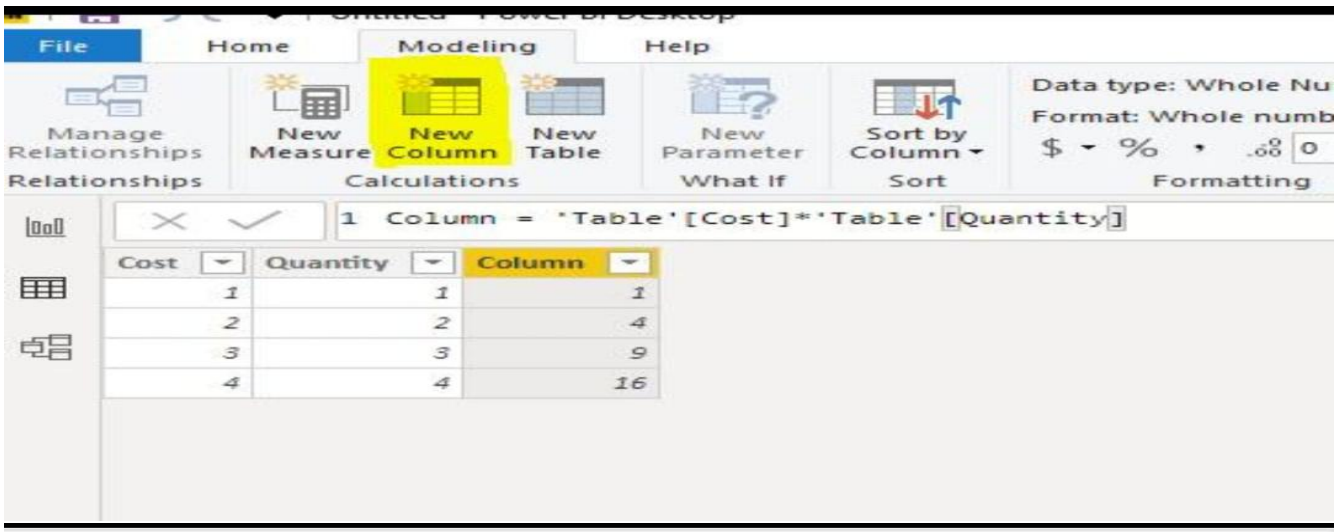

Maximum Sales = MAX(Sales[SalesAmount])

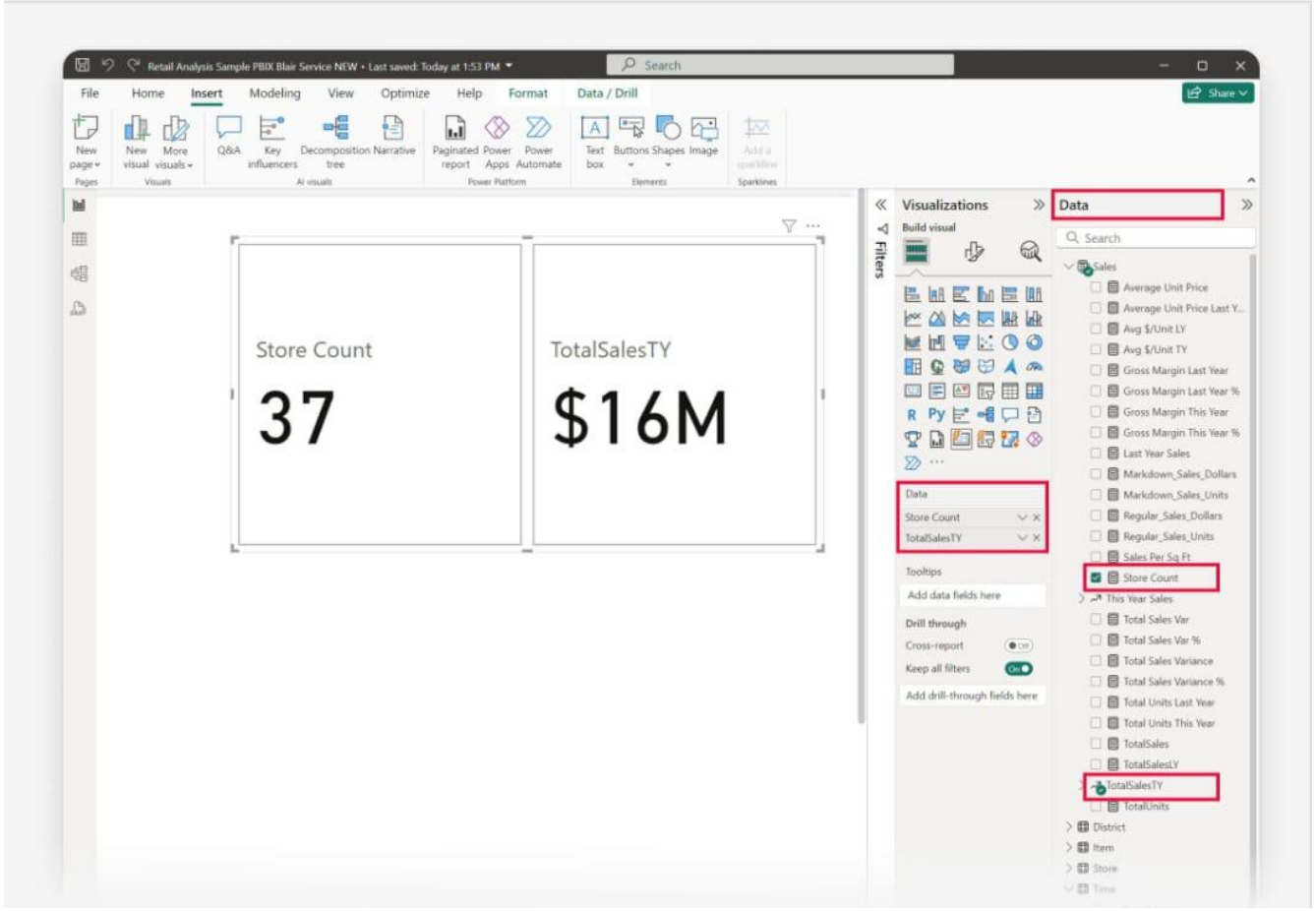

Minimum Sales = MIN(Sales[SalesAmount])


Sales Percentage = DIVIDE([Total Sales], CALCULATE([Total Sales],
ALL(Sales)), 0)


Calculated Amount = Sales[Quantity] * Sales[UnitPrice]
```

Measure / calculation	Output
Total Sales	■34,600
Average Sales	■1,647.62
Total Records	21
Maximum Sales	■5,000
Minimum Sales	■300
Calculated Amount total	■34,600





Result

The measures summarize total, average, maximum and minimum sales and count the cleaned records. The calculated column multiplies Quantity by UnitPrice.

TASK 5 – POWER BI VISUALIZATIONS

Aim

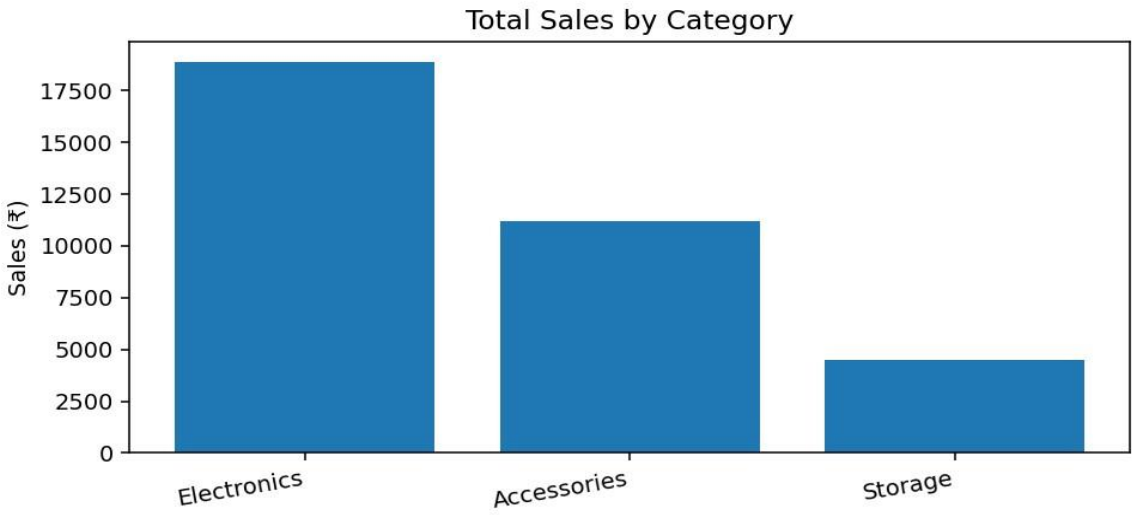
To present sales data using charts, KPI cards and a table.

Visual creation steps

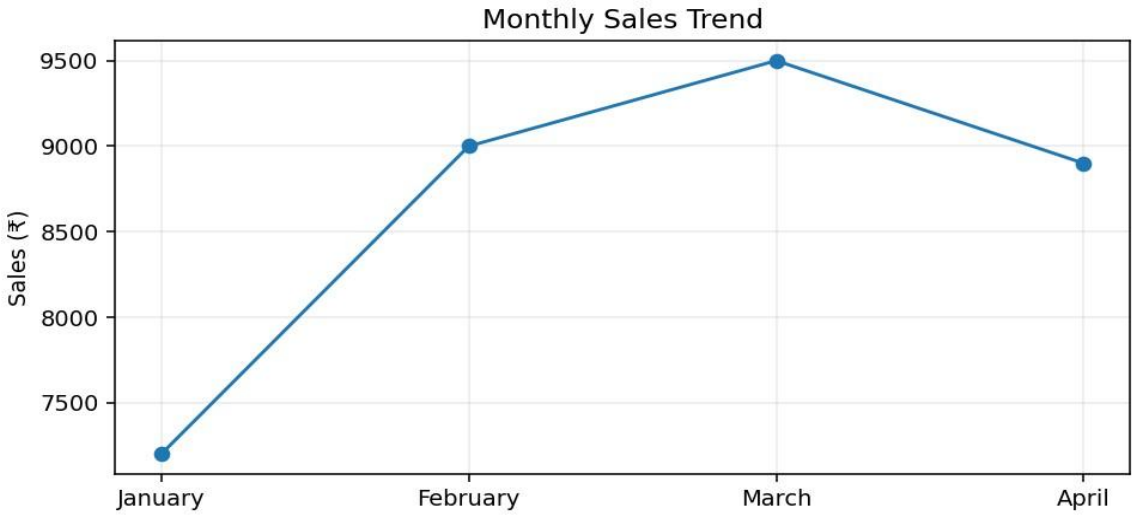
1. Add a Clustered Bar Chart: Axis = Products[Category], Values = Sum of Sales[SalesAmount].

2. Add a Line Chart: Axis = OrderDate by Month, Values = Sum of SalesAmount.
3. Add a Donut Chart: Legend = Sales[Region], Values = Sum of SalesAmount.
4. Add a Table with OrderID, OrderDate, Region, Quantity and SalesAmount.
5. Add Card visuals for Total Sales, Average Sales and Total Records.

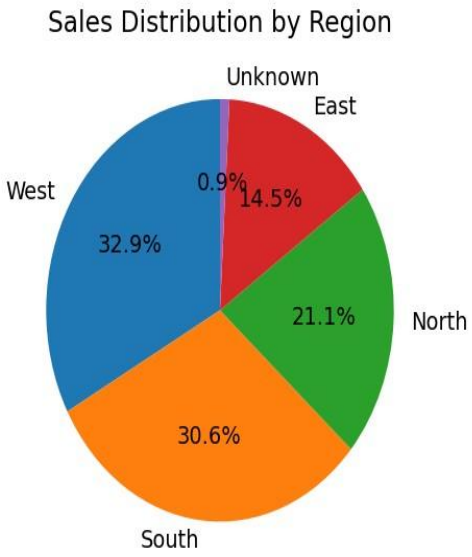
Category Sales



Monthly Sales Trend



Regional Sales Distribution



Category totals

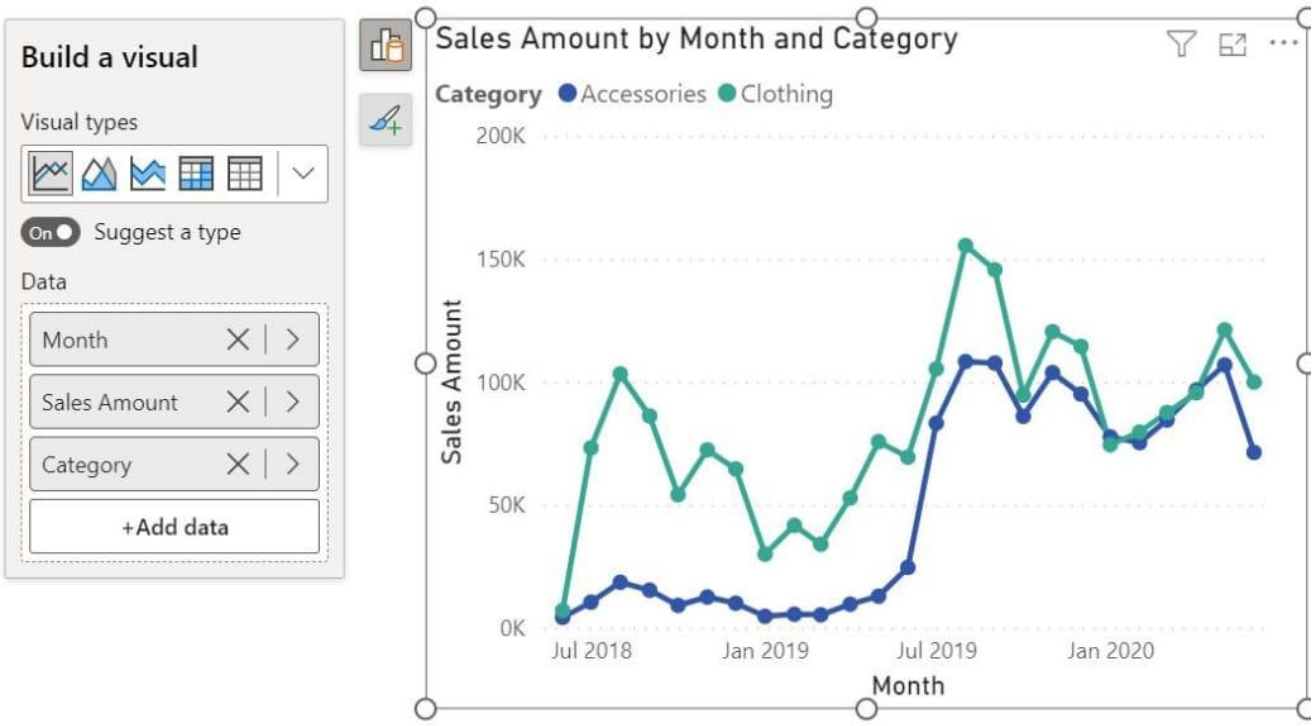
Category	Sales
Electronics	■18,900
Accessories	■11,200
Storage	■4,500

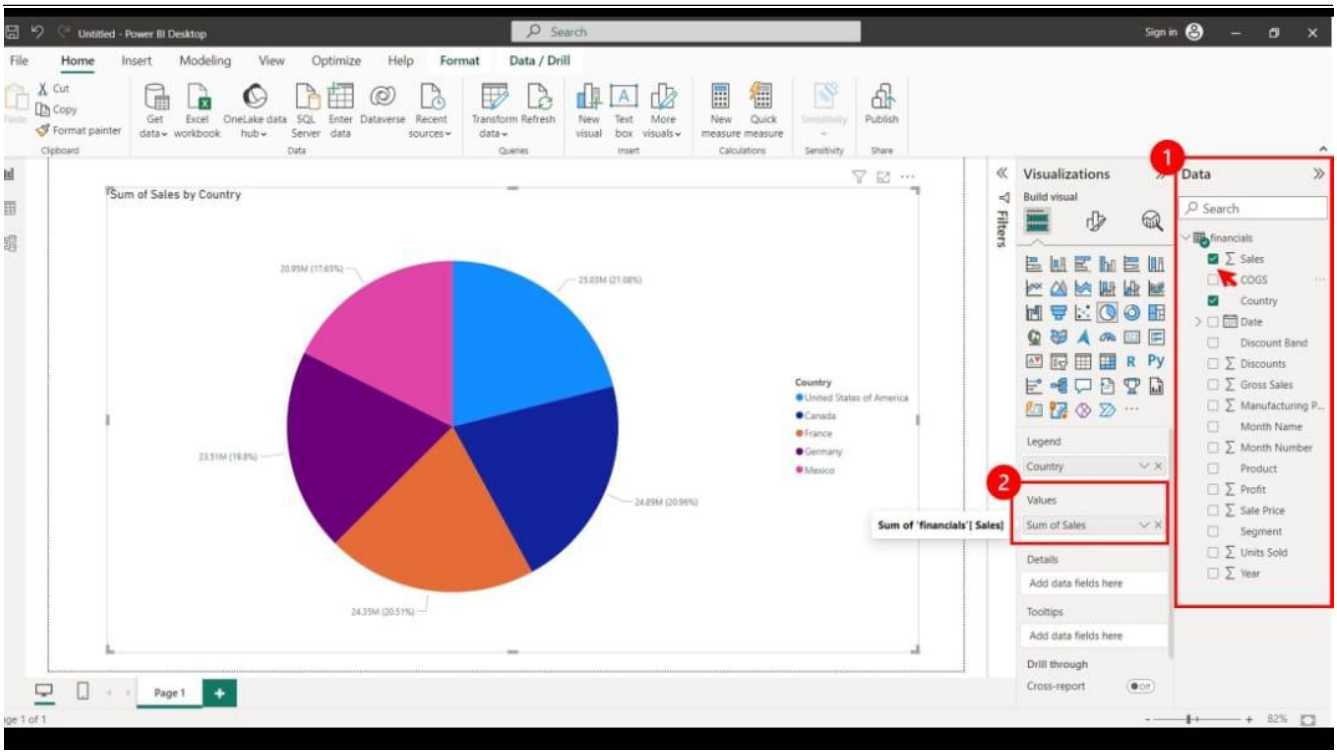
Region totals

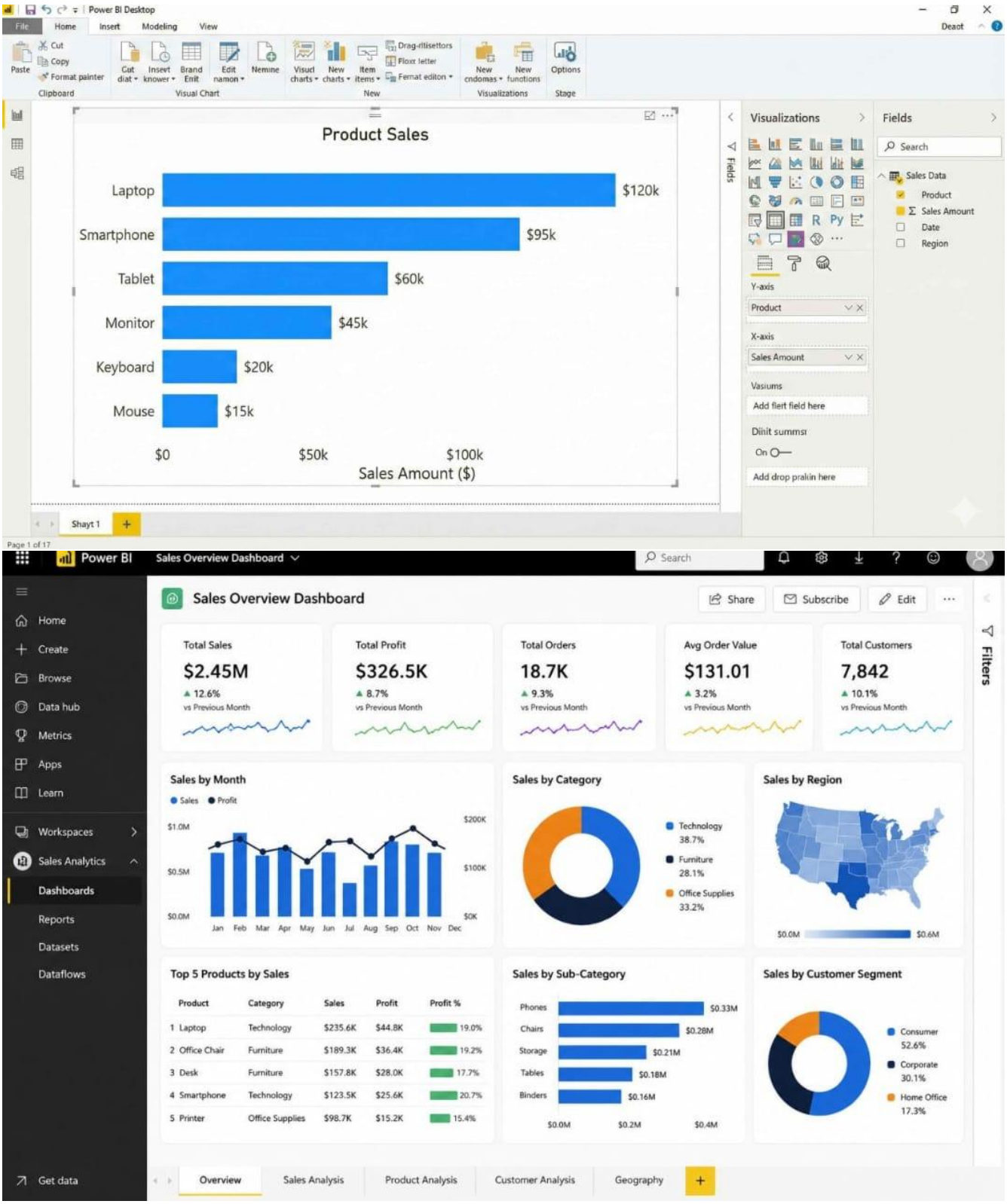
Region	Sales
West	■11,400
South	■10,600
North	■7,300
East	■5,000
Unknown	■300

Result

The visuals show category-wise, monthly and regional sales performance.


Build a visual
Visual types
On
Suggest a type
Data
Month
Sales Amount
Category
+Add data
Sales Amount by Month and Category
Category
Accessories
Clothing
200K
150K
100K
50K
0K
Sales Amount
Jul 2018
Jan 2019
Jul 2019
Jan 2020
Month


Untitled - Power BI Desktop
Search
Sign in
File
Home
Insert
Modeling
View
Optimize
Help
Format
Data / Drill
Sum of Sales by Country
Visualizations
Build visual
Data
Search
financials
Sales
COGS
Country
Date
Discount Band
Discounts
Gross Sales
Manufacturing P...
Month Name
Month Number
Product
Profit
Sale Price
Segment
Units Sold
Year
Filters
Legend
Country
Values
Sum of Sales
Details
Add data fields here
Tooltips
Add data fields here
Drill through
Cross-report
Off
Sum of 'financials'[Sales]
Country
United States of America
Canada
France
Germany
Mexico
Page 1
1
2


Power BI Desktop
Product Sales
Laptop
$120k
Smartphone
$95k
Tablet
$60k
Monitor
$45k
Keyboard
$20k
Mouse
$15k
$0
$50k
$100k
Sales Amount ($)
Visualizations
Fields
Sales Data
Product
Sales Amount
Date
Region
Y-axis
X-axis
Shayt 1
Power BI
Sales Overview Dashboard
Search
Share
Subscribe
Edit
Home
Create
Browse
Data hub
Metrics
Apps
Learn
Workspaces
Sales Analytics
Dashboards
Reports
Datasets
Dataflows
Get data
Total Sales
$2.45M
12.6%
vs Previous Month
Total Profit
$326.5K
8.7%
Total Orders
18.7K
9.3%
Avg Order Value
$131.01
3.2%
Total Customers
7,842
10.1%
Filters
Sales by Month
Sales by Category
Technology
38.7%
Furniture
28.1%
Office Supplies
33.2%
Sales by Region
Top 5 Products by Sales
Product
Category
Sales
Profit
Profit %
1 Laptop
Technology
$235.6K
$44.8K
19.0%
2 Office Chair
Furniture
$189.3K
$36.4K
19.2%
3 Desk
Furniture
$157.8K
$28.0K
17.7%
4 Smartphone
Technology
$123.5K
$25.6K
20.7%
5 Printer
Office Supplies
$98.7K
$15.2K
15.4%
Sales by Sub-Category
Phones
$0.33M
Chairs
$0.28M
Storage
$0.21M
Tables
$0.18M
Binders
$0.16M
Sales by Customer Segment
Consumer
52.6%
Corporate
30.1%
Home Office
17.3%
Overview
Sales Analysis
Product Analysis
Customer Analysis
Geography



TASK 6 – INTERACTIVE DASHBOARD & FILTERS

Aim

To combine KPIs, charts and slicers into one interactive dashboard.

Dashboard procedure

1. Add a new report page and rename it Sales Dashboard.
2. Add Card visuals for Total Sales, Average Sales and Total Records.
3. Add a Region slicer using Sales[Region].
4. Add a Category slicer using Products[Category].
5. Add the category bar chart, monthly line chart and regional donut chart.
6. Add a sales details table.
7. Test the slicers and confirm that visuals update.
8. Use Format → Edit interactions if needed.
9. Save as PowerBI_Sales_Analysis.pbix.

Illustrative dashboard layout

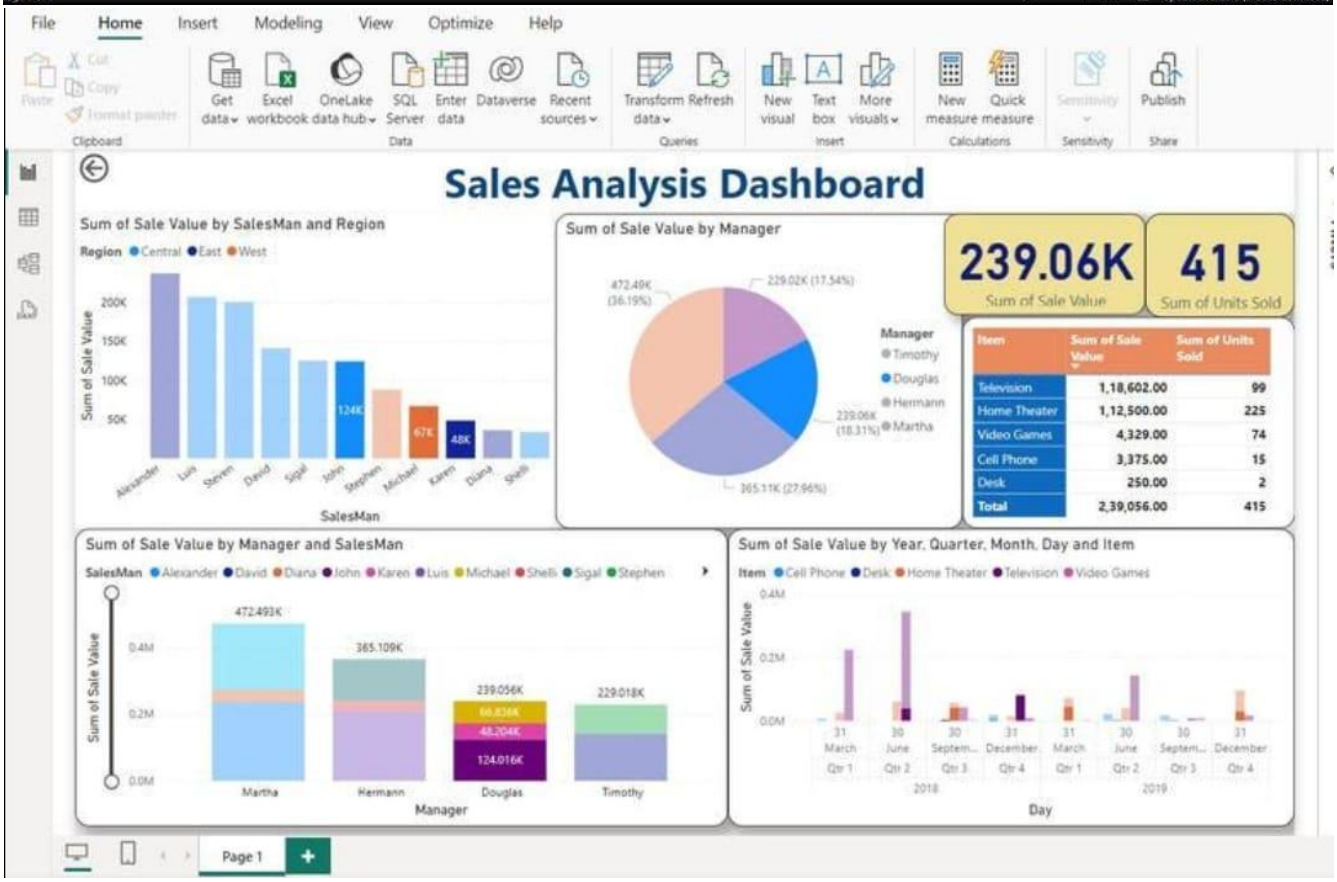
KPI card 1	KPI card 2	KPI card 3
Total Sales ■34,600	Average Sales ■1,647.62	Total Records 21
Region slicer	Category slicer	
Category chart	Monthly trend chart	
Regional chart	Sales details table	

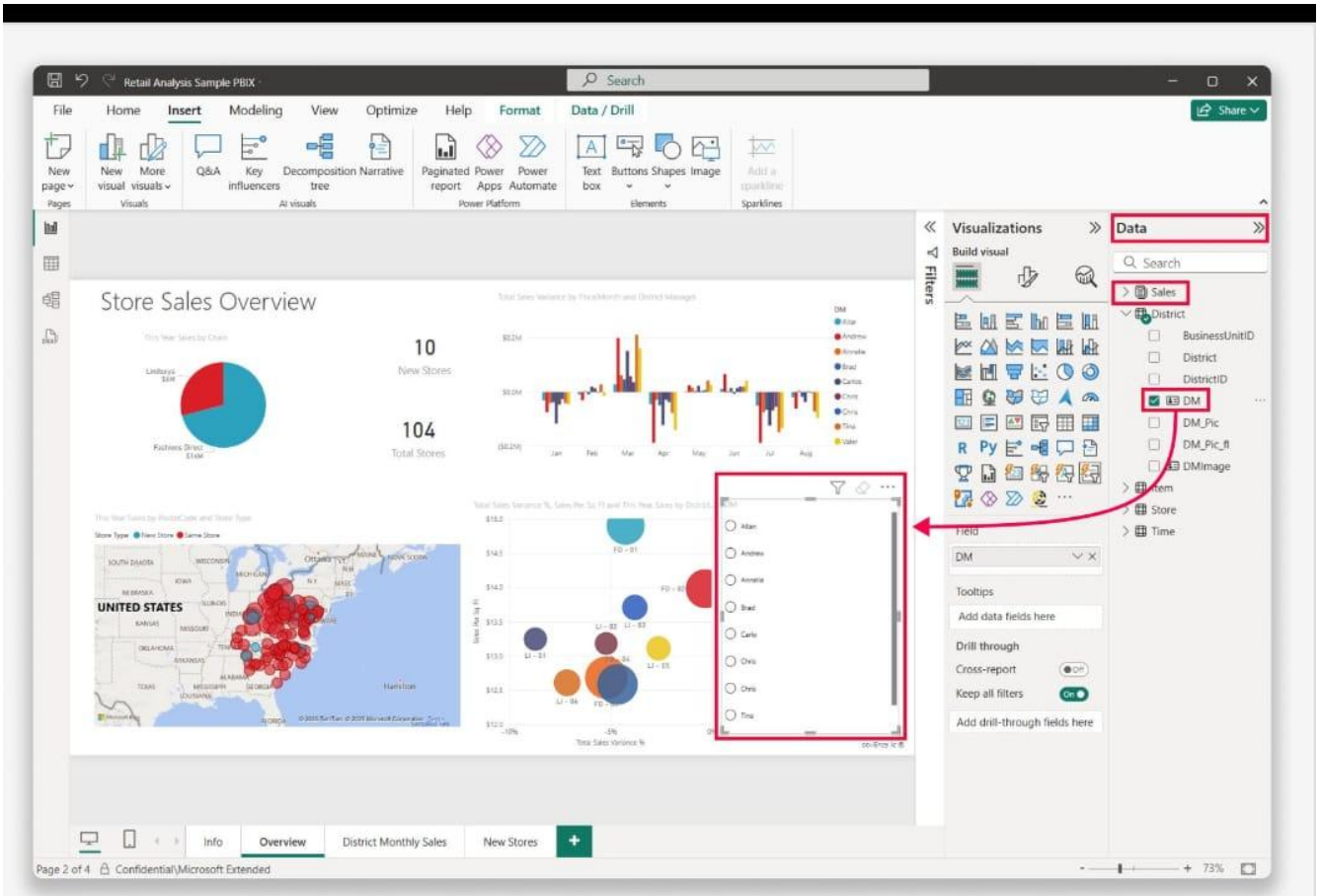
Screenshot note: The layout is a guide, not an actual dashboard screenshot. Capture your completed dashboard in Power BI Desktop and insert it before submission if required.

Result

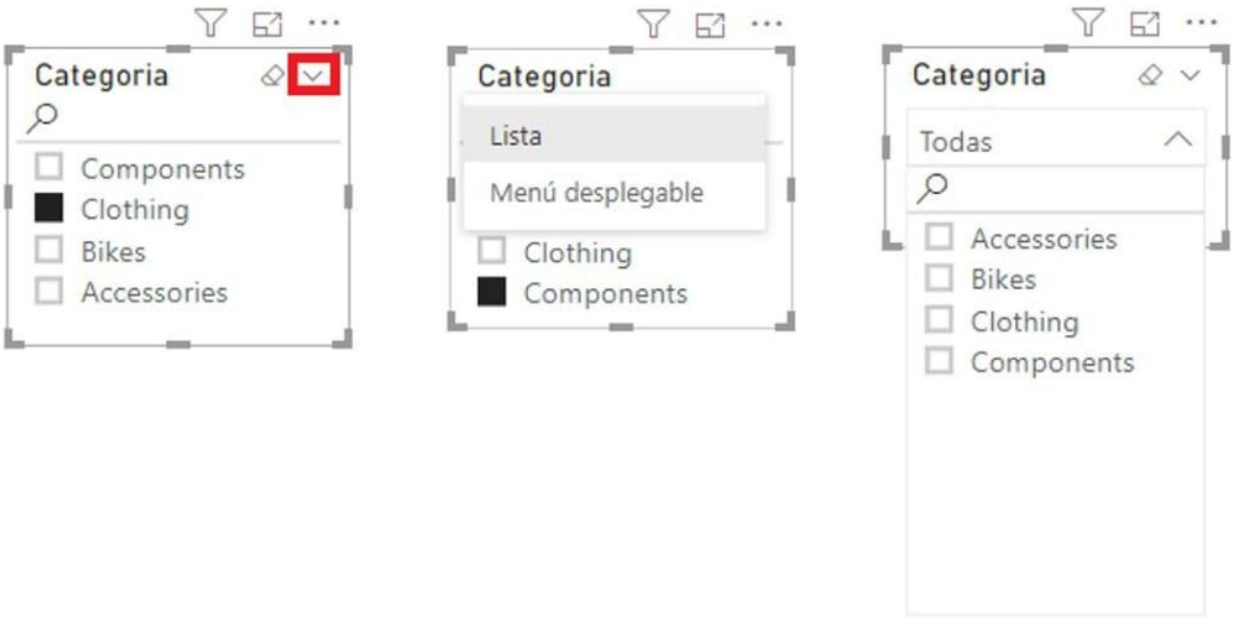
The dashboard contains three KPI cards, two slicers, charts and a transaction details table.


Sales Summary
Sales Details
Monthly performance
Total No of Orders
540
Total Sales Amounts
0.30M
Total No of products
10
Total No of Sales Person
3
Total Sales by Continent
Sum of Sales by Date
Total No of Orders by Team
Count of Order ID by Country and Continent
Date
1/19/2022
12/2/2024
Sales Person
Country
Select all
Audio
Computing
Electronics
Summary
More Details
Sales Details
Monthly performance
©UHDNM JAYALATH


Sales Analysis Dashboard
Sum of Sale Value by SalesMan and Region
Sum of Sale Value by Manager
239.06K
Sum of Sale Value
415
Sum of Units Sold
Item
Sum of Sale Value
Sum of Units Sold
Television 1,18,602.00 99
Home Theater 1,12,500.00 225
Video Games 4,329.00 74
Cell Phone 3,375.00 15
Desk 250.00 2
Total 2,39,056.00 415
Sum of Sale Value by Manager and SalesMan
Sum of Sale Value by Year, Quarter, Month, Day and Item
Page 1


Retail Analysis Sample PBIX
Search
File
Home
Insert
Modeling
View
Optimize
Help
Format
Data / Drill
Share
Store Sales Overview
10
New Stores
104
Total Stores
UNITED STATES
Visualizations
Build visual
Data
Search
Sales
District
BusinessUnitID
District
DistrictID
DM
DM_Pic
DM_Pic_fl
DMImage
Item
Store
Time
Filters
Field
DM
Tooltips
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Info
Overview
District Monthly Sales
New Stores
Page 2 of 4
Confidential\Microsoft Extended
73%


Total Sales
33.93M
Total Quantity Sold
86K
Average Sales
1.44K
Number of Orders
23.603K
Filter by Product
Product Name
AWC Logo Cap
Bike Wash - Dissolver
Cable Lock
Chain
Sales by Date
Total Sales
2M
0M
2012
2013
2014
Date
Sales by Region
2.28M (6.72%)
4.83M (14.22%)
11.94M (35.19%)
5.2M (15.32%)
6.94M (20.45%)
Territory
Canada
Northwest
France
United Kingdom
Filter by year
Year
2011
2012
2013
2014
Filter by Region
Territory
Australia
Canada
Central
France
Orders Statues
1.85K (7.82%)
2.65K (11.21%)
6.37K (26.99%)
5.69K (24.11%)
6.05K (25.63%)
Status
Approved
In process
Shipped
Cancelled
Rejected
Backordered
Top 5 Product
100%
Mountain-20...
1.53M
Mountain-20...
1.21M
Mountain-20...
1.03M
Mountain-20...
0.96M
Touring-1000...
0.90M
58.4%
Sales by product
2.0M
1.5M
1.0M
0.5M
0.0M
Total sales
700
800
900
Product ID



TASK 7 – BUSINESS INSIGHTS & RECOMMENDATIONS

Aim

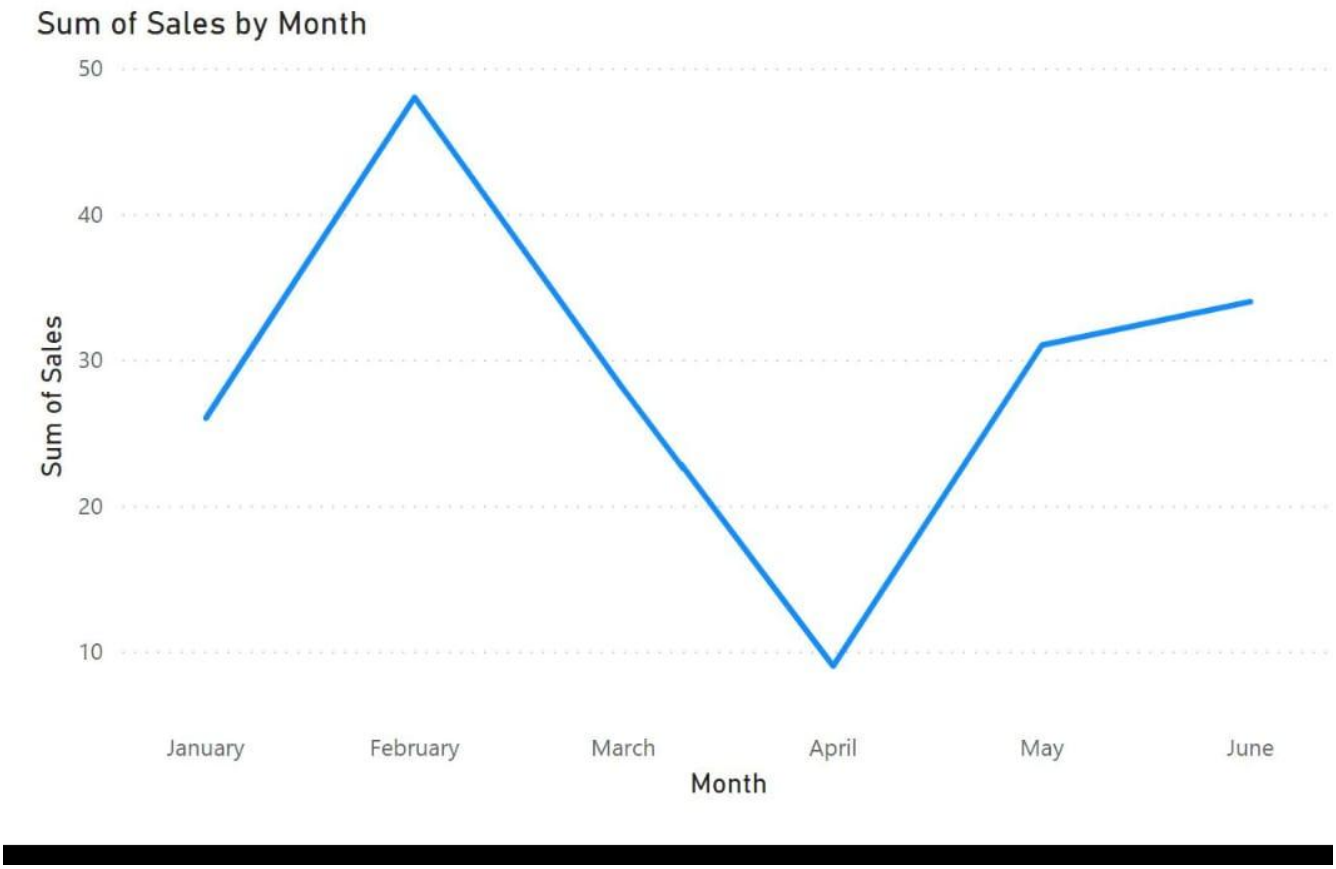
To analyze the powber Bi dashboard,identify important business trends,evaluate category-wise and region-wise performance, and provide business recommandations

Procedure

1.Analyze total sales
2.Analyze category wise sales
3.Analyze region-wise sales
4.Analyze monthly sales







Business Insights

1. Electronics is the highest-selling category at ■18,900.
2. Storage is the lowest-selling category at ■4,500.
3. West is the highest-performing region at ■11,400.

4. East contributes ■5,000; Unknown contributes a separate ■300.
5. Monthly sales rise from ■7,200 in January to ■9,500 in March, then decrease slightly to ■8,900 in April.
6. Total sales are ■34,600 across 21 records, with average sales of ■1,647.62.

Recommendations

• Continue promoting Electronics products.
• Study customer demand and consider targeted offers for Storage products.
• Review customer preferences in the East region and plan suitable promotions.
• Monitor the April decline and plan marketing activities accordingly.
• Keep product, customer and transaction data accurate and up to date.

Conclusion

This project demonstrates the Power BI workflow from data import and cleaning to modeling, DAX calculations, visualization, dashboard design and business interpretation. The cleaned dataset contains 21 records and total sales of ■34,600. The report highlights product, regional and monthly sales performance.